

Supplementary Materials: Self-Organized cancer fragility emerges from chromosomal instability

Theoretical Biology & Mathematical Oncology Framework

June 16, 2026

1 Stationary Mutational Population distributions in the Liquid case

We consider a liquid tumor model with fixed population size N . The number of wild-type cells is indicated with W , the number of dead cells with D . The cancer cell population can be further divided in N_I sub-populations (one per instability class), with N_I the number of instability genes. We indicate the abundance of each cancer sub-population with C_k , where $k \in \{1, \dots, N_I\}$ counts the number of mutated instability genes. We work with densities (fractions) rather than absolute abundances:

$$f_W \equiv \frac{W}{N}, \quad f_k \equiv \frac{C_k}{N}, \quad p_D \equiv \frac{D}{N} \quad (1)$$

These are constrained by the fixed-size normalization:

$$\sum_{k=1}^{N_I} f_k + f_W + p_D = 1 \quad (2)$$

Survival probability after a mutation round

Both cancer and wild-type cells undergo replication followed by a stochastic mutation event. After mutation, a *viability check* is performed on all N_{HK} housekeeping genes: if *any* housekeeping gene becomes fully mutated (on all chromosome copies), the cell dies. Each gene has an independent per-locus mutation probability $\mu_k = k \cdot \delta\mu$ for a class- k cell. The probability that a *single* housekeeping gene survives (i.e. is **not** mutated) in one round is $(1 - k \delta\mu)$. Since all N_{HK} housekeeping genes must independently survive:

$$P_{s,k} = \underbrace{(1 - k \delta\mu)}_{\text{one HK gene survives}} \times \dots \times \underbrace{(1 - k \delta\mu)}_{N_{HK} \text{ times}} = (1 - k \delta\mu)^{N_{HK}} \quad (3)$$

The post-replication fates (exact multi-step model)

After a mutation round, a cell in class k can either remain in class k (cloning), promote to class $k + j$ with $j \geq 1$ (multi-step promotion), or die due to a housekeeping mutation. Since mutations at different loci occur independently with per-locus probability $\mu_k = k \cdot \delta\mu$, we derive these probabilities exactly:

1. Faithful cloning (α_k): The cell remains in class k . This requires that none of the N_{HK} housekeeping genes and none of the $(N_I - k)$ remaining unmutated instability genes are hit. The total number of loci that must remain unmutated is $N_{HK} + N_I - k$:

$$\alpha_k = (1 - k \delta\mu)^{N_{HK} + N_I - k} \quad (4)$$

2. Promotion ($P_{k \rightarrow k+j}$): The cell advances from class k to class $k+j$ (where $j \in \{1, \dots, N_I - k\}$ instability genes mutate, while all other instability loci and all housekeeping loci remain unmutated). Since the selection of which instability genes mutate is combinatorial:

$$P_{k \rightarrow k+j} = \binom{N_I - k}{j} (k \delta \mu)^j (1 - k \delta \mu)^{N_{HK} + N_I - k - j} \quad (5)$$

Note that faithful cloning is simply the diagonal case $\alpha_k = P_{k \rightarrow k}$.

3. Lethal mutation (δ_k): The cell dies because at least one housekeeping gene is hit, which is the complement of the survival probability $P_{s,k} = (1 - k \delta \mu)^{N_{HK}}$:

$$\delta_k = 1 - P_{s,k} = 1 - (1 - k \delta \mu)^{N_{HK}} \quad (6)$$

Exact normalization check: The sum of all possible survival outcomes plus the death probability is exactly 1:

$$\sum_{j=0}^{N_I - k} P_{k \rightarrow k+j} + \delta_k = (1 - k \delta \mu)^{N_{HK}} \sum_{j=0}^{N_I - k} \binom{N_I - k}{j} (k \delta \mu)^j (1 - k \delta \mu)^{N_I - k - j} + [1 - (1 - k \delta \mu)^{N_{HK}}] = 1 \quad (7)$$

This binomial formulation provides an exact description of the mutation process without needing leading-order approximations.

Moran competition probabilities

When a dividing cell targets a living (non-dead) cell, the outcome is decided by a *Moran process*: the divider replaces the target with probability proportional to its fitness relative to the combined fitness. All cancer classes share the same intrinsic division rate $r_{\max}$, while wild-type cells divide at rate r_0 :

- **Cancer vs Wild-Type:** The cancer cell (rate $r_{\max}$) competes against a wild-type cell (rate r_0):

$$\beta_{cw} = \frac{r_{\max}}{r_{\max} + r_0} \quad (8)$$

- **Cancer vs Cancer:** Two cancer cells with the same rate $r_{\max}$ compete symmetrically:

$$\beta_{cc} = \frac{r_{\max}}{r_{\max} + r_{\max}} = \frac{1}{2} \quad (9)$$

- **Wild-Type vs Cancer:** The complement of the cancer-vs-WT outcome:

$$\beta_{wc} = \frac{r_0}{r_{\max} + r_0} = 1 - \beta_{cw} \quad (10)$$

System parameters are formalized in Table 1.

1.1 Microscopic Mechanisms and Elementary Transition Rates

After each duplication step, both mother and daughter cells independently undergo stochastic mutation events as described above. Both cells independently draw from the same probability distribution: cloning (α_k), promoting ($P_{k \rightarrow k+j}$), or dying (δ_k). Crucially, DNA replication and subsequent viability/mutation checks are only performed upon successful division (Moran competition success). If division is aborted, neither the mother nor the daughter replicates or mutates. Thus, all division-dependent rates (cloning, promotion, and death) for both mother and daughter cells are scaled by the division success probability ϕ_k .

Table 1: Systematic Overview of the Unified Parameter Space

Symbol	Biological Definition	Functional Form / Constraint
N	Total fixed tissue lattice sites	Constant volume constraint
N_I	Maximum viable cancer mutational classes	Total mutable Instability genes
N_{HK}	Total critical Housekeeping genes	Continuous fitness check loci
$\delta\mu$	Per-locus mutation rate increment	Constant scaling factor
$r_{\max}$	Intrinsic division rate of class k cancer	Phenotypic fitness parameter
r_0	Active division rate of healthy wild-type	Homeostatic driver baseline
α_k	Probability of a faithful clone	$(1 - k \delta\mu)^{N_{HK}+N_I-k}$
$P_{k \rightarrow k+j}$	Probability of promotion ($k \rightarrow k+j$)	$\binom{N_I-k}{j} (k \delta\mu)^j (1 - k \delta\mu)^{N_{HK}+N_I-k-j}$
δ_k	Lethal mutation probability	$1 - (1 - k \delta\mu)^{N_{HK}}$
β_{cw}	Moran win probability of cancer vs WT	$r_{\max}/(r_{\max} + r_0)$
β_{cc}	Moran win probability of cancer k vs j	$r_{\max}/(r_{\max} + r_{\max})$
β_{wc}	Moran win probability of WT vs cancer k	$r_0/(r_{\max} + r_0) = 1 - \beta_{cw}$

Effective reachable space ϕ_k

We define the fractional reachable space ϕ_k as the effective probability that a dividing class- k cancer cell successfully places its daughter. The daughter can land on three types of targets:

$$\phi_k = \underbrace{p_D}_{\text{dead slots: always available}} + \underbrace{f_W \beta_{cw}}_{\text{WT cells displaced via Moran competition}} + \underbrace{\sum_{j=1}^{N_I} f_j \beta_{cc}}_{\text{cancer cells displaced (50% win rate)}} \quad (11)$$

- p_D : the fraction of dead (empty) slots. Replacement always succeeds (no competition needed).
- $f_W \beta_{cw}$: wild-type cells are encountered with frequency f_W , and the cancer cell wins the Moran competition with probability β_{cw} .
- $\sum_j f_j \beta_{cc}$: other cancer cells are encountered with total frequency $\sum_j f_j$, with a symmetric 50% win rate $\beta_{cc} = 1/2$.

Enumeration of all elementary transitions

We now enumerate every possible microscopic event and its rate. The key insight is that during a single division event of a class- k cancer cell, *two independent fates* are resolved: (i) the **mother fate** (what happens to the original cell on its site after mutation), and (ii) the **daughter fate** (what happens to the copy placed at the target site after mutation). Both fates are conditioned on the division being successfully completed.

The complete transition rate matrix is given in Table 2. We explain each row below.

Explanation of each transition rate:

1. **Mother dies on site** ($C_k r_{\max} \phi_k \delta_k$): A class- k cell successfully initiates division (probability ϕ_k), and the mother undergoes a lethal housekeeping mutation (probability δ_k). The cell dies in place: $C_k \rightarrow C_k - 1$, $D \rightarrow D + 1$.
2. **Mother promotes on site** ($C_k r_{\max} \phi_k P_{k \rightarrow k+j}$): The mother successfully initiates division and mutates j instability genes, advancing to class $k+j$ *without leaving its site*. This is a type-swap: $C_k \rightarrow C_k - 1$, $C_{k+j} \rightarrow C_{k+j} + 1$, with no change to D .

Table 2: Complete Microscopic Transition Rate Matrix (Symmetrical Hazards)

Active Vector	Target & Event Type	ΔC_k	ΔD	Transition Rate (λ)
Cancer Class k	Mother Fate: Dies on Site	-1	+1	$C_k r_{\max} \phi_k \delta_k$
Cancer Class k	Mother Fate: Promotes on Site ($k \rightarrow k+j$)	$-1_k, +1_{k+j}$	0	$C_k r_{\max} \phi_k P_{k \rightarrow k+j}$
Cancer Class k	Daughter Fate: Clones on D	+1	-1	$C_k r_{\max} p_D \alpha_k$
Cancer Class k	Daughter Fate: Promotes on D ($k \rightarrow k+j$)	$+1_{k+j}$	-1	$C_k r_{\max} p_D P_{k \rightarrow k+j}$
Cancer Class k	Daughter Fate: Dies on D	0	0	$C_k r_{\max} p_D \delta_k$
Cancer Class k	Daughter Fate: Clones on W	+1	0	$C_k r_{\max} f_W \beta_{cw} \alpha_k$
Cancer Class k	Daughter Fate: Dies on W	0	+1	$C_k r_{\max} f_W \beta_{cw} \delta_k$
Wild-Type (W)	WT Divides, Replaces Dead Void	0	-1	$W r_0 p_D$
Wild-Type (W)	WT Divides, Clears Cancer k	-1	0	$W r_0 f_k \beta_{wc}$

3. **Daughter clones on dead slot** ($C_k r_{\max} p_D \alpha_k$): The daughter targets a dead slot (probability p_D) and faithfully clones (probability α_k). Net effect: $C_k \rightarrow C_k + 1$, $D \rightarrow D - 1$.
4. **Daughter promotes on dead slot** ($C_k r_{\max} p_D P_{k \rightarrow k+j}$): The daughter targets a dead slot and mutates j instability genes during division. Net effect: $C_{k+j} \rightarrow C_{k+j} + 1$, $D \rightarrow D - 1$.
5. **Daughter dies on dead slot** ($C_k r_{\max} p_D \delta_k$): The daughter targets a dead slot but suffers a lethal mutation. The dead slot remains dead, and no net population change occurs ($\Delta C_k = 0$, $\Delta D = 0$).
6. **Daughter clones on WT** ($C_k r_{\max} f_W \beta_{cw} \alpha_k$): The daughter targets a wild-type cell (probability f_W), wins the Moran competition (probability β_{cw}), and faithfully clones (α_k). The WT cell is displaced: $C_k \rightarrow C_k + 1$, $W \rightarrow W - 1$.
7. **Daughter dies on WT** ($C_k r_{\max} f_W \beta_{cw} \delta_k$): The daughter wins competition against a WT cell but then suffers a lethal mutation. The WT cell is lost and a dead cell appears: $D \rightarrow D + 1$, $W \rightarrow W - 1$.
8. **WT replaces dead slot** ($W r_0 p_D$): A wild-type cell divides (rate r_0) and fills a dead slot (probability p_D). Net: $W \rightarrow W + 1$, $D \rightarrow D - 1$. No Moran competition is needed against dead cells.
9. **WT clears cancer** ($W r_0 f_k \beta_{wc}$): A wild-type cell targets a class- k cancer cell (probability f_k) and wins the Moran competition with probability β_{wc} . Net: $C_k \rightarrow C_k - 1$, $W \rightarrow W + 1$.

1.2 Mean-Field Ordinary Differential Equations

By taking the thermodynamic limit ($N \rightarrow \infty$), the stochastic master equation converges to a deterministic system of ODEs. We derive the ODE for each compartment by summing all transition rates from Table 2 that affect that compartment, converting from absolute counts to densities ($f_k = C_k/N$).

Derivation of df_k/dt for interior cancer classes ($1 < k < N_I$)

For an interior cancer class k ($1 < k < N_I$), the rate of change of the cell fraction f_k is determined by five primary biological processes: (i) net clonal growth and survival ($r_{\max} f_k \phi_k (2\alpha_k - 1)$), which balances faithful daughter cloning against mother cell loss via death or promotion; (ii) self-displacement loss within the same class ($r_{\max} f_k^2/2$); (iii) cross-class loss due to displacement by other cancer classes ($f_k \sum_{j \neq k} r_{\max} f_j \beta_{cc}$); (iv) wild-type clearance loss ($r_0 f_W f_k \beta_{wc}$); and

(v) promotion influx from all upstream classes $i < k$ ($\sum_{i=1}^{k-1} 2 r_{\max} f_i \phi_i P_{i \rightarrow k}$). Combining these contributions yields:

$$\frac{df_k}{dt} = r_{\max} f_k \phi_k (2\alpha_k - 1) - r_{\max} \frac{f_k^2}{2} - f_k \sum_{j \neq k} r_{\max} f_j \beta_{cc} - r_0 f_W f_k \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i \phi_i P_{i \rightarrow k} \quad (12)$$

Boundary class $k = 1$

The founding cancer class $k = 1$ receives **no upstream mutational influx** (there is no class $k = 0$). The equation is identical to the interior case but without the promotion influx:

$$\frac{df_1}{dt} = r_{\max} f_1 \phi_1 (2\alpha_1 - 1) - r_{\max} \frac{f_1^2}{2} - f_1 \sum_{j \neq 1} r_{\max} f_j \beta_{cc} - r_0 f_W f_1 \beta_{wc} \quad (13)$$

Terminal class $k = N_I$

For the terminal mutational class, all instability genes are already mutated, so further promotion is impossible: $P_{N_I \rightarrow N_I+j} = 0$. The mother-loss term simplifies to $-r_{\max} f_{N_I} \phi_{N_I} (1 - \alpha_{N_I})$ where $1 - \alpha_{N_I} = \delta_{N_I}$. The promotion influx from all classes $i < N_I$ is present:

$$\frac{df_{N_I}}{dt} = r_{\max} f_{N_I} \phi_{N_I} (2\alpha_{N_I} - 1) - r_{\max} \frac{f_{N_I}^2}{2} - f_{N_I} \sum_{j \neq N_I} r_{\max} f_j \beta_{cc} - r_0 f_W f_{N_I} \beta_{wc} + \sum_{i=1}^{N_I-1} 2 r_{\max} f_i \phi_i P_{i \rightarrow N_I} \quad (14)$$

Wild-type compartment f_W

The WT population grows when WT cells divide and successfully place daughters on dead slots or displace cancer cells, and shrinks when cancer daughters displace WT cells:

- **Gain:** WT divides onto dead slots (rate $r_0 f_W p_D$) or displaces cancer class j (rate $r_0 f_W f_j \beta_{wc}$ per class).
- **Loss:** Cancer class k daughters displace WT cells (rate $r_{\max} f_k f_W \beta_{cw}$ per class).

$$\frac{df_W}{dt} = \underbrace{r_0 f_W \left(p_D + \sum_{j=1}^{N_I} f_j \beta_{wc} \right)}_{\text{WT expansion (dead slots + cancer clearance)}} - \underbrace{f_W \sum_{k=1}^{N_I} r_{\max} f_k \beta_{cw}}_{\text{displacement by cancer daughters}} \quad (15)$$

Dead compartment p_D

Dead cells accumulate from lethal mutations (both mother dying in place and daughter dying after placement) and are consumed when any living cell fills a dead slot. Since DNA replication only happens during successful division, both mother and daughter can only die if division succeeds.

- **Gain from mother death:** $r_{\max} f_k \phi_k \delta_k$ per class (row 1).
- **Gain from daughter death:** $r_{\max} f_k \phi_k \delta_k$ per class (combining rows 5, 7, and cancer-cancer targets).
- **Total death-related gain:** $2 r_{\max} f_k \phi_k \delta_k$ per class.

- **Loss from filling:** Dead slots are consumed by cancer daughters ($r_{\max} f_k p_D$, regardless of fate) and WT daughters ($r_0 f_W p_D$).

$$\frac{dp_D}{dt} = \underbrace{\sum_{k=1}^{N_I} r_{\max} f_k [2\phi_k \delta_k - p_D]}_{\text{cancer-driven death accumulation minus filling}} - \underbrace{r_0 f_W p_D}_{\text{WT filling dead slots}} \quad (16)$$

1.3 Stationary State Distributions and Stability Analysis

Linear stability of the healthy fixed point

We examine the linear stability of the healthy fixed point ($f_W^* = 1$, $p_D^* = 0$, $f_k^* = 0 \forall k$) against an infinitesimal perturbation $f_1 = \epsilon \ll 1$, with all other cancer classes still at zero.

Starting from the $k = 1$ boundary ODE and substituting the healthy fixed-point values, each term simplifies as follows:

- **Mother loss & daughter clone (combined):** $r_{\max} \epsilon \phi_1 (2\alpha_1 - 1)$. At the fixed point, $\phi_1 \rightarrow f_W \beta_{cw} = 1 \cdot \beta_{cw}$. This gives $r_{\max} \beta_{cw} (2\alpha_1 - 1) \epsilon$.
- **Self-displacement:** $-r_{\max} \epsilon^2 / 2 \approx 0$ — vanishes at $\mathcal{O}(\epsilon^2)$.
- **Cross-class loss:** $-\epsilon \sum_{j \neq 1} r_{\max} f_j \beta_{cc} = 0$.
- **WT clearance:** $-r_0 (1) \epsilon \beta_{wc} = -r_0 \beta_{wc} \epsilon$.
- **Promotion influx:** absent for $k = 1$.

Collecting all $\mathcal{O}(\epsilon)$ terms:

$$\frac{df_1}{dt} \approx [r_{\max} \beta_{cw} (2\alpha_1 - 1) - r_0 \beta_{wc}] \epsilon \quad (17)$$

Substituting $\beta_{cw} = r_{\max} / (r_{\max} + r_0)$ and $\beta_{wc} = r_0 / (r_{\max} + r_0)$:

$$\frac{df_1}{dt} \approx \frac{r_{\max}}{r_{\max} + r_0} [r_{\max} (2\alpha_1 - 1) - r_0] f_1 \quad (18)$$

Derivation of the growth foothold inequality

The tumor can invade when the bracketed growth rate is positive:

$$r_{\max} (2\alpha_1 - 1) - r_0 > 0 \quad (19)$$

This yields the **Unified Growth Foothold Inequality**:

$$\boxed{r_{\max} (2\alpha_1 - 1) > r_0} \quad (20)$$

Physical interpretation: The left-hand side balances the cancer cell's maximum birth rate times its net clonal expansion factor ($2\alpha_1 - 1$), which accounts for the survival and fidelity of both the mother and daughter during a division event. The right-hand side r_0 represents the WT homeostatic division pressure. When mutation rates are too high, the cancer's net growth rate drops below the wild-type's division rate, preventing the tumor from taking foothold.

Expression in terms of the survival probability $P_{s,1}$

We can rewrite the inequality in terms of the survival probability $P_{s,1} = (1 - \delta\mu)^{N_{HK}}$ and the probability of faithful cloning $\alpha_1 = P_{s,1}(1 - \delta\mu)^{N_I-1}$:

$$r_{\max} \left[2P_{s,1} (1 - \delta\mu)^{N_I-1} - 1 \right] > r_0 \quad (21)$$

This form makes the phase boundary explicit: the critical $\delta\mu^*$ is the value at which $P_{s,1}$ (which decreases with $\delta\mu$) drops low enough to violate the inequality.

Stationary state: reduction to sequential quadratic equations

When the growth inequality is satisfied, the system reaches a tumor-dominated stationary state where $df_k/dt = 0$ for all k . Setting the interior ODE to zero and rearranging:

$$0 = r_{\max} f_k^* \phi_k^* (2\alpha_k - 1) - r_{\max} \frac{(f_k^*)^2}{2} - f_k^* \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \quad (22)$$

We separate the ϕ_k^* sum into its self-interaction part ($f_k^* \beta_{cc}$) and the rest ($\phi_k^{(\neq k)*}$):

$$\phi_k^* = \phi_k^{(\neq k)*} + f_k^* \beta_{cc} \quad (23)$$

Substituting and grouping terms by powers of f_k^* :

$$\begin{aligned} 0 &= r_{\max} f_k^* \left[\phi_k^{(\neq k)*} + f_k^* \beta_{cc} \right] (2\alpha_k - 1) - r_{\max} \frac{(f_k^*)^2}{2} \\ &\quad - f_k^* \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \\ 0 &= r_{\max} f_k^* \phi_k^{(\neq k)*} (2\alpha_k - 1) + r_{\max} (f_k^*)^2 \beta_{cc} (2\alpha_k - 1) - r_{\max} \frac{(f_k^*)^2}{2} \\ &\quad - f_k^* \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \end{aligned} \quad (24)$$

Since $\beta_{cc} = 1/2$, the quadratic terms combine as:

$$-r_{\max} (1 - \alpha_k) (f_k^*)^2 \quad (25)$$

Collecting terms, this takes the standard quadratic form:

$$A_k (f_k^*)^2 - B_k f_k^* - Q_{k-1}^* = 0 \quad (26)$$

where the coefficients are:

A_k — **Quadratic (self-interaction) coefficient:**

$$A_k = r_{\max} (1 - \alpha_k) \quad (27)$$

This arises from the combination of self-displacement ($\beta_{cc} = 1/2$) and the mutation outcomes.

B_k — **Linear (net growth) coefficient:**

$$B_k = r_{\max} \phi_k^{(\neq k)*} (2\alpha_k - 1) - \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* \beta_{wc} \quad (28)$$

This coefficient represents the *net per-capita growth rate* of class k at the stationary state, excluding self-interaction and upstream promotion.

Q_{k-1}^* — **Upstream promotion source:**

$$Q_{k-1}^* = \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \quad (29)$$

This is the rate at which cells flow into class k from all lower classes $i < k$ (dual-sourced: mother promoting + daughter promoting). For $k = 1$, $Q_0^* = 0$.

Solution via the quadratic formula

The quadratic $A_k x^2 - B_k x - Q_{k-1}^* = 0$ (with $x = f_k^*$) has the physical (positive) root:

$$f_k^* = \frac{B_k + \sqrt{B_k^2 + 4 A_k Q_{k-1}^*}}{2 A_k} \quad (30)$$

The positive sign before the square root is chosen because $f_k^* \geq 0$ is required. The system is solved *sequentially*: f_1^* is determined first (with $Q_0^* = 0$), then f_2^* using f_1^* , and so on up to $f_{N_I}^*$.